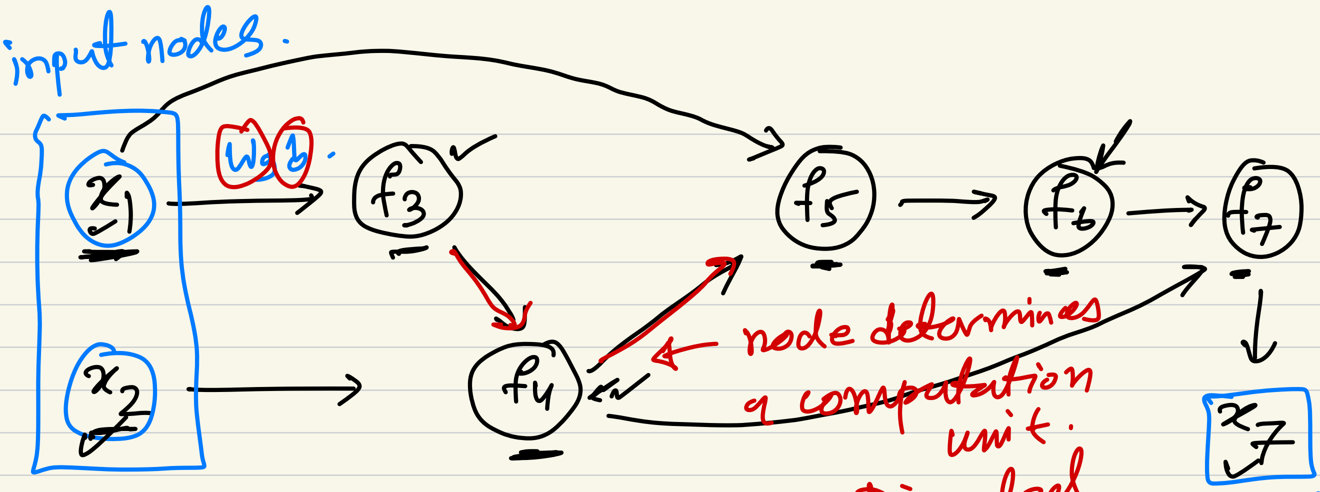


forward propagation

computation graph. ↳ Should be DAG

Directed acyclic graph

output node.

↓

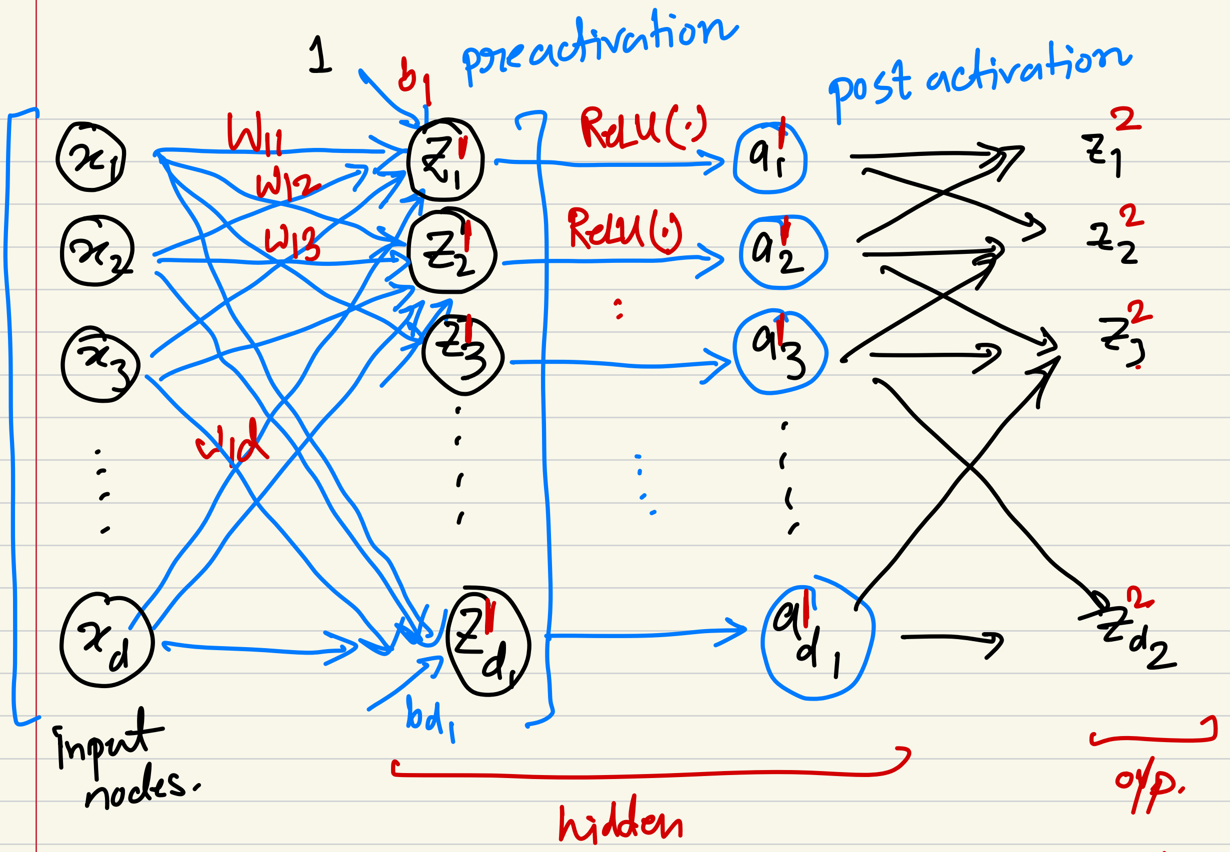
L

$$
\begin{cases}
f_3 = 2x_1 + 5 = w_1 x_1 + b. & = 2/x_1 \times 5 \\
f_4 = 3x_2 + 6 + f_3 & = \log x_2 + 5 \\
f_5 = 5x_1 + 6f_4 + 3 & \\
f_6 = 6f_5 + 5 & \\
f_7 = 3f_4 + 5f_6 + 8 & \\
x_7 = 3f_7 + 5 &
\end{cases}
$$

$$\boxed{L = -\log x_7}$$

$$L = -y \log(\hat{y}) - (1-y)\log(1-\hat{y})$$

$$L = (x_7 - y)^2$$



$$z_1^1 = W_{11}\cdot x_1 + W_{12}\cdot x_2 + \cdots + W_{1d}\cdot x_d + b_1^1$$

$$z_2^1 = W_{21}x_1 + W_{22}x_2 + \cdots + W_{2d}x_d + b_2^1$$

$$\vdots$$

$$z_{d_1}^1 = W_{d_1\cdot 1}x_1 + W_{d_1\cdot 2}x_2 + \cdots + W_{d_1\cdot d}x_d + b_{d_1}^1$$

$\Downarrow$ matrix form.

$$\underbrace{\begin{bmatrix} z_1^1 \\ z_2^1 \\ \vdots \\ z_{d_1}^1 \end{bmatrix}}_{\in \mathbb{R}^{d_1}} = \underbrace{\begin{bmatrix} W_{11} & W_{12} & \cdots & W_{1d} \\ W_{21} & W_{22} & \cdots & W_{2d} \\ & \vdots & & \\ W_{d_1,1} & W_{d_1\cdot 2} & \cdots & W_{d_1,d} \end{bmatrix}}_{\in \mathbb{R}^{d_1 \times d}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}}_{\in \mathbb{R}^{d}} + \underbrace{\begin{bmatrix} b_1^1 \\ b_2^1 \\ \vdots \\ b_{d_1}^1 \end{bmatrix}}_{\in \mathbb{R}^{d_1}}$$

$$z_1 = W_1 \cdot x + b_1$$

$$x \longrightarrow \underset{W_1, b_1}{\boxed{\text{Linear}}} \longrightarrow z_1$$

$$a_1^1 = \text{ReLU}(z_1^1) = \max(0, z_1^1)$$

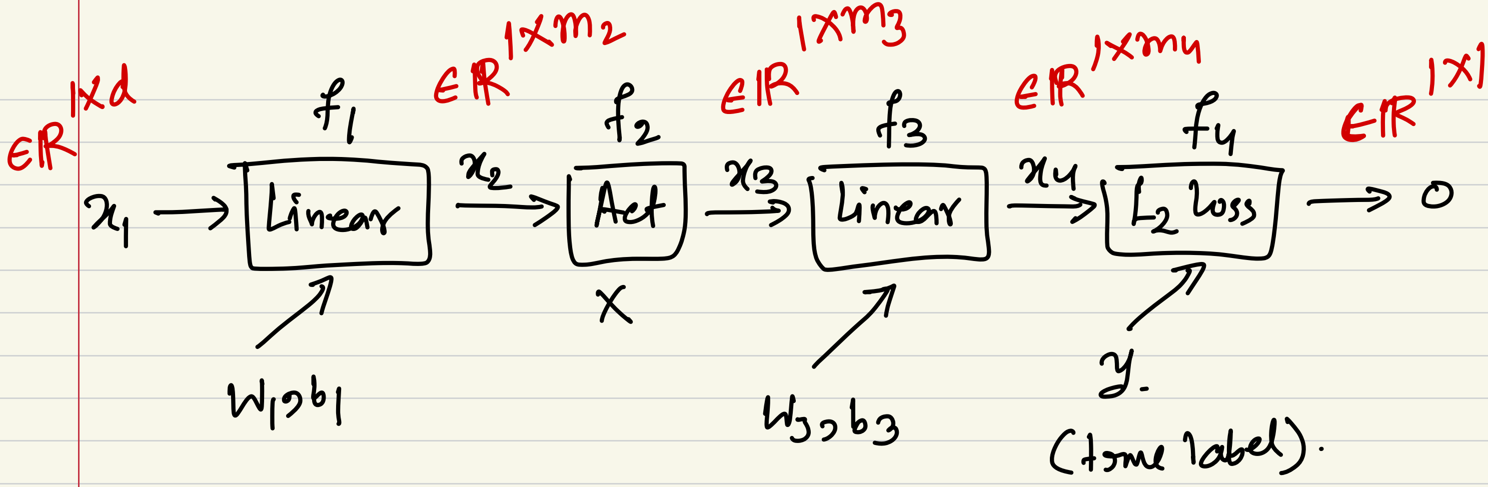
$$a_2^1 = \text{ReLU}(z_2^1) = \max(0, z_2^1)$$

$$\vdots$$

$$a_{d_1}^1 = \text{ReLU}(z_{d_1}^1) = \max(0, z_{d_1}^1)$$

$$\Downarrow$$

$$\begin{bmatrix} a_1^1 \\ \vdots \\ a_{d_1}^1 \end{bmatrix} = \underbrace{\text{ReLU}}_{\text{elementwise}} \left(\begin{bmatrix} z_1^1 \\ z_2^1 \\ z_{d_1}^1 \end{bmatrix} \right)$$



computation graph.

$$x_2 = x_1 \cdot W_1 + b_1$$

$$x_3 = act(x_2)$$

$$x_4 = x_3 \cdot W_3 + b_3$$

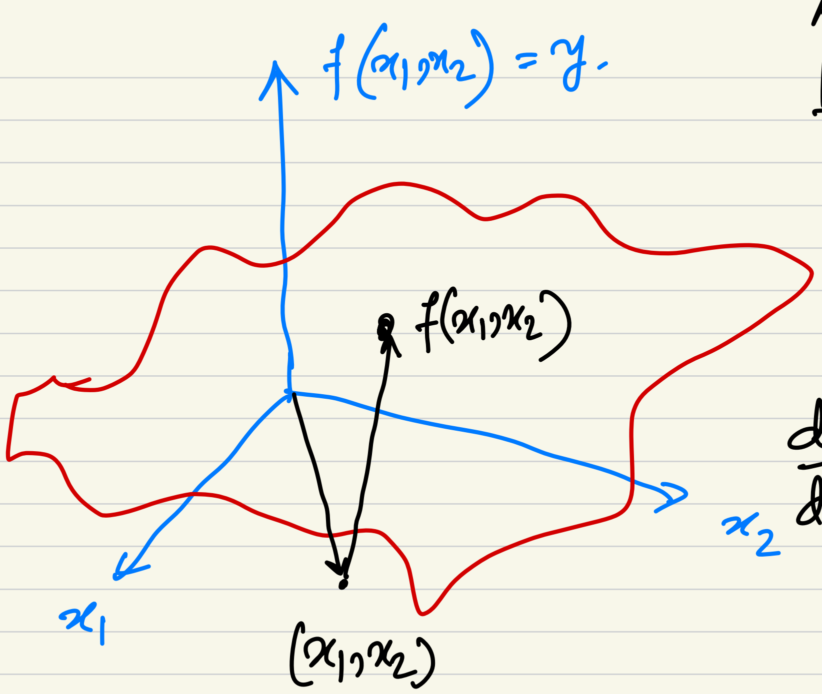
$$o = f_4(y, x_4) = -\underbrace{y \log x_4}.$$

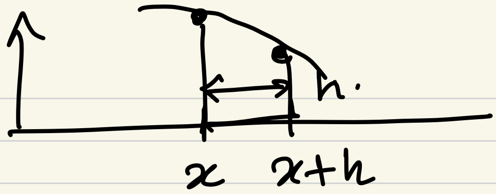
$$[\;] \cdot \log([\;])$$

Backpropagation is the process of computing the gradient of loss with respect to the parameters of the model.

Chain rule of calculus.

$$y = f(x_1, x_2)$$





$$\frac{df}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

What is the rate of change of function in the direction of x_1 and x_2?

$$\frac{\partial f}{\partial x_1}, \quad \frac{\partial f}{\partial x_2}.$$

$$y = \underbrace{(2x_1 + 3x_2)}^{2} = (z)^2 \qquad y = z^2$$

$$z = 2x_1 + 3x_2$$

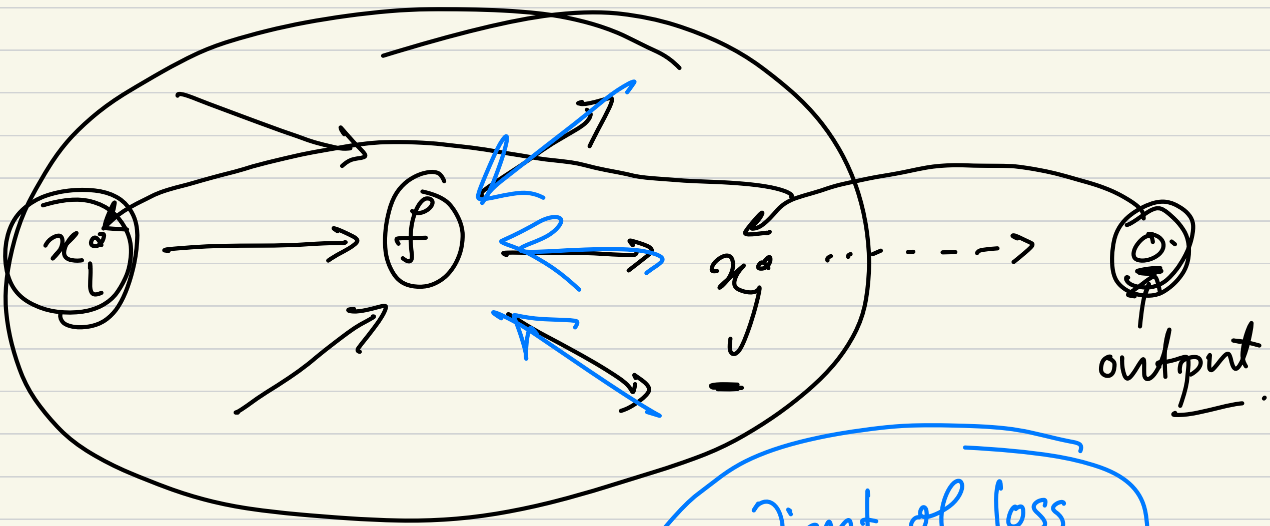
$$\frac{\partial y}{\partial x_1} = 2(2x_1+3x_2).(2) \qquad \frac{\partial y}{\partial z} = 2z$$

$$\frac{\partial y}{\partial x_2} = 2(2x_1+3x_2).(3). \qquad \frac{\partial y}{\partial x_1}, \frac{\partial y}{\partial x_2} =$$

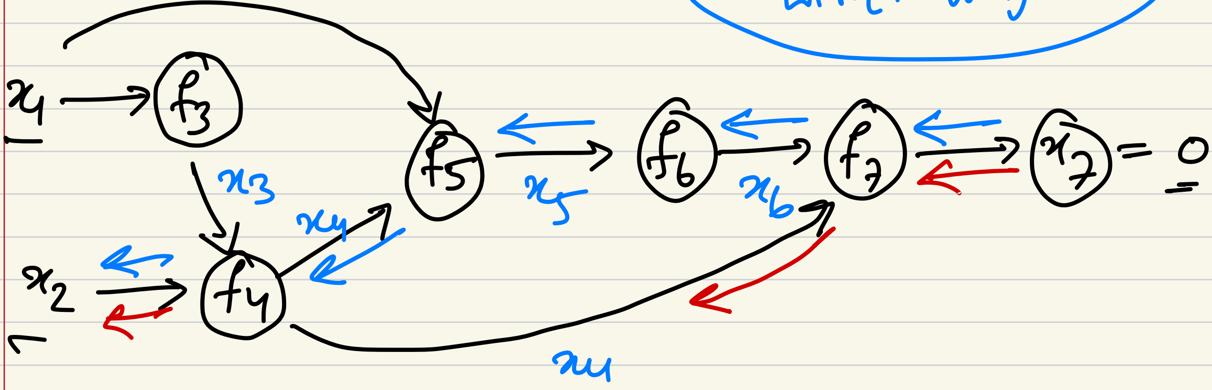
$$\boxed{\frac{\partial y}{\partial x_1} = \frac{\partial y}{\partial z} \cdot \frac{\partial z}{\partial x_1}} +$$

$$\frac{\partial y}{\partial x_2} = \frac{\partial y}{\partial z} \cdot \frac{\partial z}{\partial x_2}$$

chain rule of calculus.



gradient of loss w.r.t. weights



$$\frac{\partial O}{\partial x_2} = ?$$

gradient contribution from path:

$$x_7 \to x_6 \to x_5 \to x_4 \to x_2$$

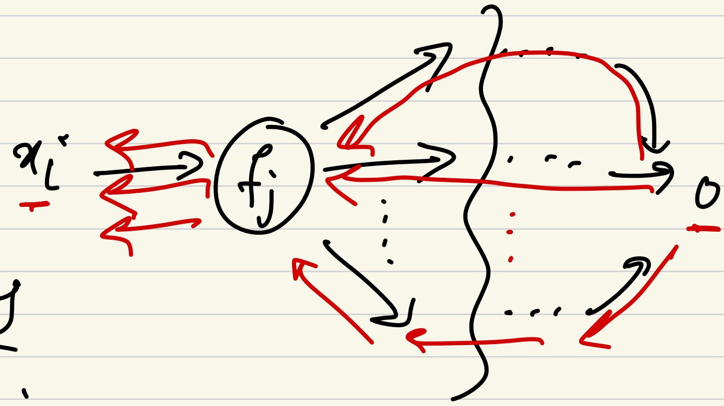
$$\frac{\partial O}{\partial x_2} = \boxed{\frac{\partial O}{\partial x_6} \cdot \frac{\partial x_6}{\partial x_5} \cdot \frac{\partial x_5}{\partial x_4}} \cdot \frac{\partial x_4}{\partial x_2}.$$

$$= \frac{\partial O}{\partial x_4} \cdot \frac{\partial x_4}{\partial x_2}.$$

gradient contribution from path.

$$x_7 \rightarrow x_4 \rightarrow x_2.$$

$$\frac{\partial o}{\partial x_2} = \frac{\partial o}{\partial x_4} \cdot \frac{\partial x_4}{\partial x_2}.$$



$$\frac{\partial o}{\partial x_i} = \sum_{j=1}^{c} \frac{\partial o}{\partial x_j} \cdot \frac{\partial x_j}{\partial x_i}$$

c number of childrens

c = number of childrens of node f_j

$$x \longrightarrow \boxed{W_1, b_1} \longrightarrow z_1 \longrightarrow \boxed{W_2, b_2} \longrightarrow z_2$$

$$\boxed{z_1 = W_1 x + b_1} \longrightarrow \frac{\partial z_1}{\partial W_1} = x$$

$$\boxed{z_2 = W_2 \cdot z_1 + b_2} \longrightarrow \frac{\partial z_2}{\partial z_1} = W_2$$

$$\frac{\partial z_2}{\partial W_1} = \frac{\partial z_2}{\partial z_1} \cdot \frac{\partial z_1}{\partial W_1} + (\quad) + (\ldots) = W_2 \cdot x.$$

1. Forward prop. : $Z = Wx + b$.

2. Back prop : $\frac{\partial Z}{\partial W}, \frac{\partial Z}{\partial b}$

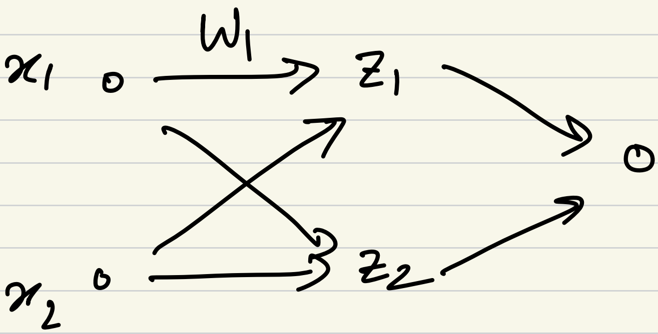
3. optimization :

$$\left[\begin{array}{l} W_{new} \leftarrow W_{old} - \eta \cdot \frac{\partial Z}{\partial W} \\ b_{new} \leftarrow b_{old} - \eta \cdot \frac{\partial Z}{\partial b} \end{array}\right]$$

$$x \longrightarrow W_1, b_1 \longrightarrow z_1 \longrightarrow W_2, b_2 \longrightarrow z_2 \longrightarrow W_3, b_3 \longrightarrow z_3 \longrightarrow L$$

$$\begin{bmatrix} \frac{\partial L}{\partial W_1} & \frac{\partial L}{\partial b_1} \\ \frac{\partial L}{\partial W_2} & \frac{\partial L}{\partial b_2} \\ \frac{\partial L}{\partial W_3} & \frac{\partial L}{\partial b_3} \end{bmatrix}$$

$W_1 \in \mathbb{R}^{m \times n}$.



$$\frac{\partial O}{\partial W_1} = ?$$